

5-Point Correlation Function in the 2D Ising CFT

1. Units of the Quantities

In the context of a Conformal Field Theory (CFT) defined on a 2D plane or cylinder, the fundamental "unit" is the length dimension, denoted as L . The operators (fields) and correlation functions have specific scaling dimensions with respect to L .

- **Positions (x_i):** The coordinates x_i (representing the complex positions z_i or their real equivalents) have the dimension of **Length**, $[x_i] = L$.
- **Correlation Function ($\langle \dots \rangle$):** The value of the correlation function scales with the inverse of the total scaling dimension of the operators involved.
 - The Energy operator ϵ has scaling dimension $\Delta_\epsilon = 1$ (1/2 holomorphic + 1/2 anti-holomorphic).
 - The Spin operator σ has scaling dimension $\Delta_\sigma = 1/8$ (1/16 holomorphic + 1/16 anti-holomorphic).
 - The 5-point function $\langle \epsilon \epsilon \epsilon \sigma \sigma \rangle$ has a total scaling dimension of:

$$[\langle \epsilon \epsilon \epsilon \sigma \sigma \rangle] = 3\Delta_\epsilon + 2\Delta_\sigma = 3(1) + 2(1/8) = 3.25$$

- Thus, the correlation function has dimensions of $L^{-3.25}$.

2. Results of Dimensional Analysis

We analyzed the derived formula for the correlation function using the dimensional analysis tool.

Formula:

$$G = \frac{1}{2} \cdot |x_{45}|^{-1/4} \cdot \sqrt{\frac{|x_{15}| |x_{25}| |x_{35}|}{|x_{14}| |x_{24}| |x_{34}|}} \cdot \frac{1}{|x_{12}| |x_{23}| |x_{31}|}$$

Tool Input:

- Equation: `G = 0.5 * x_45^(-0.25) * sqrt((x_15 * x_25 * x_35) / (x_14 * x_24 * x_34)) * (1 / (x_12 * x_23 * x_31))`
- Dimensions: `{"G": "correlation", "x_ij": "length"}`
- Unit List: `length`

Tool Output:

The dimensional analysis resulted in `length**-3.25`.

Analysis:

The output dimensions match the theoretically expected scaling dimension of the 5-point correlation function ($L^{-3.25}$). Therefore, the formula is dimensionally consistent.

3. Corrected Mathematical Description

Based on the consistency check and the derivation using the fermionic representation of the Ising model, the final mathematical model for the 5-point function is:

$$\langle \epsilon(x_1)\epsilon(x_2)\epsilon(x_3)\sigma(x_4)\sigma(x_5) \rangle = \frac{1}{2} \frac{\sqrt{|x_{15}||x_{25}||x_{35}|}}{|x_{45}|^{1/4} \sqrt{|x_{14}||x_{24}||x_{34}|} \cdot |x_{12}||x_{23}||x_{31}|}$$

Computation Results

Case (1): Complex Coordinates

- $x_1 = 1 + i, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5$

$$\langle \epsilon\epsilon\epsilon\sigma\sigma \rangle = \frac{\sqrt{3} \cdot 17^{1/4}}{2\sqrt{2} \cdot 10^{3/4}} \approx 0.312$$

Case (2): Real Coordinates

- $x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5$

$$\langle \epsilon\epsilon\epsilon\sigma\sigma \rangle = \frac{1}{2}$$